

David Puentes

Full-Stack Software Developer · Data Analyst · IoT & Applied AI Builder

Bogotá, Colombia · github.com/DavidSPuentesc

PROFILE

Engineer-minded developer building web products, data workflows, connected electronics, and applied AI systems. Experience spans TypeScript, React, Python, SQL, Docker, ESP32, LoRa, scientific computing, computer vision, and agentic workflows with skills, subagents, context control, token budgets, traceability, and human validation.

EXPERIENCE

Full-Stack Software Developer & Data Analyst · PRLCOL

Aug 2025—Present

- Develop and support complete modules for a multi-instance B2B SaaS platform.
- Work across SQL migrations, automation, telemetry, integrations, and applied AI.

Research & Development Developer · Universidad ECCI / CEINTECCI

Jun 2024—Jun 2025

- Built reproducible scientific-computing pipelines for stylometric analysis over approximately two million syllables.
- Supported technical communication and research dissemination.

SELECTED PROJECTS

SmartSense Monitoring

Industrial thermal monitoring with PT100, ESP32-C6, LoRa, local storage, and dashboard. More than 51,000 samples recorded; a 688 m obstructed field link was demonstrated.

SecurApp

Software and applied-AI work involving orchestration, specialized agents and subagents, reusable skills, context and token management, and human-reviewed outcomes.

Electronics, sensing & scientific systems

Projects involving Wi-Fi sensing, sensor dashboards, ESP32 firmware, computer vision, scientific simulation, and data analysis.

TECHNICAL SKILLS

AI & automation: agent orchestration, subagents, skills, RAG, prompt design, token budgeting, computer vision

Software & data: TypeScript, React, Python, Django, Flask, SQL, REST APIs, Docker

Electronics & IoT: ESP32, C++, LoRa, BLE, PT100, sensors, instrumentation, Edge AI

Infrastructure & security: Terraform, AWS, CI/CD, observability, cybersecurity and privacy

CERTIFICATIONS

AI Application Development · React.js with TypeScript · TypeScript · Terraform · Cybersecurity and Privacy for Businesses —
Platzi (2026)
Scientific Computing with Python · Responsive Web Design — freeCodeCamp (2025)